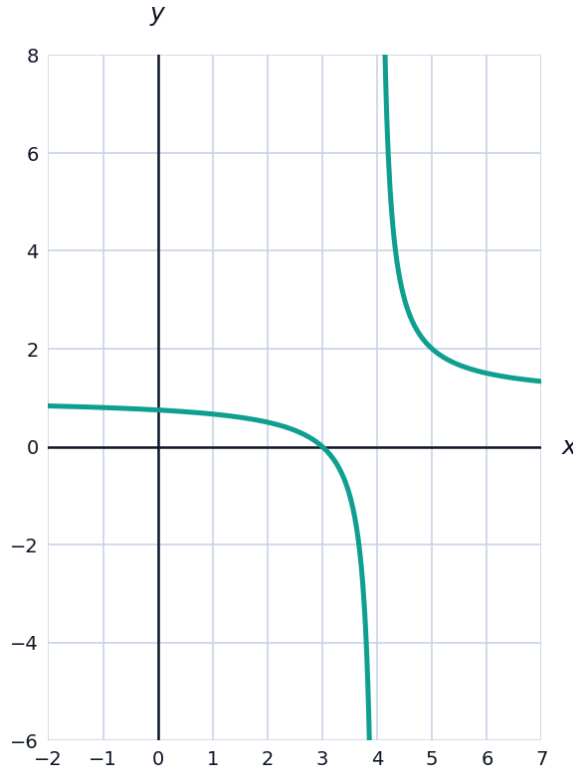


Rational Functions & Asymptotes



Vertical asymptotes and holes

- 1 Find the vertical asymptotes and holes of $f(x) = \frac{x^2 - 9}{x^2 - x - 12}$.



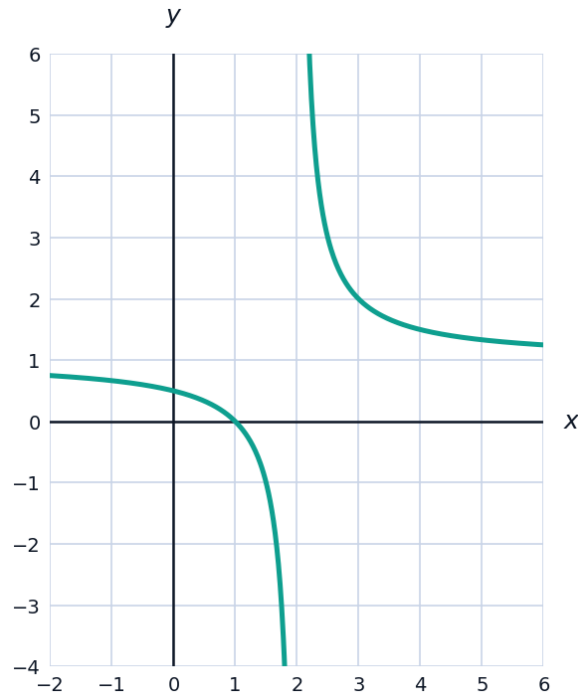
- 2 Find the y-coordinate of the hole in the function from the previous problem.

Horizontal and slant asymptotes

- 3 Find the horizontal asymptote of $f(x) = \frac{3x^2 - 2x + 1}{5x^2 + 4}$, using the degrees of numerator and denominator.
- 4 Find the horizontal asymptote of $g(x) = \frac{2x + 7}{x^2 - 1}$.
- 5 Explain why $h(x) = \frac{x^2 + 1}{x - 2}$ has a slant (oblique) asymptote instead of a horizontal one, and find it using polynomial division.

Visualizing a rational function's asymptotes

- 6 The graph shows $f(x) = \frac{1}{x-2} + 1$, a transformed reciprocal function.



- a Identify the vertical asymptote shown.
- b Identify the horizontal asymptote shown.

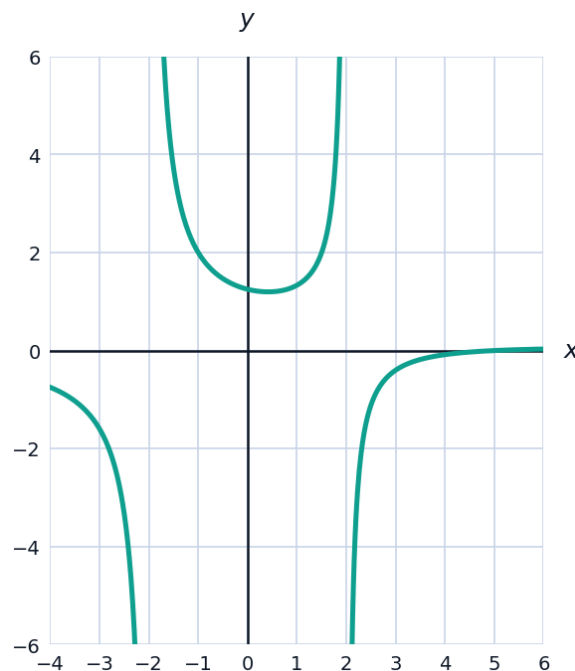
End behavior of rational functions

- 7 Describe the end behavior of $f(x) = \frac{2x^3 - x}{x^2 + 1}$ as $x \rightarrow \pm \infty$ (note the numerator's degree exceeds the denominator's by 1, so the function behaves like its slant asymptote).



Domain and intercepts

- 8 Find the domain, x-intercept(s), and y-intercept of $f(x) = \frac{x - 5}{x^2 - 4}$.



Solving rational equations and inequalities

- 9 Solve $\frac{x+1}{x-3} = 2$, checking that the solution doesn't create a zero denominator.
- 10 Solve the inequality $\frac{x-1}{x+2} > 0$ by determining the sign of the expression on each interval created by the zero and the excluded value.

Visualizing a function with a slant asymptote

- 11 The graph shows $f(x) = \frac{x^2}{x-1}$, which has a slant asymptote.



- a Find the vertical asymptote and confirm it matches the graph.
- b Use polynomial division to find the slant asymptote equation.
- c Explain why this function has no horizontal asymptote.

Building a rational function from given features

- 12 Construct a rational function with a vertical asymptote at $x = 3$, a hole at $x = -1$, and a horizontal asymptote at $y = 2$.

- 13 Explain why a rational function cannot have both a horizontal asymptote and a slant asymptote at the same time.

Rational functions in context

- 14 The concentration of a drug in the bloodstream t hours after a dose is modeled by $C(t) = \frac{50t}{t^2 + 4}$ (mg/L). Find the horizontal asymptote and interpret it in context.

